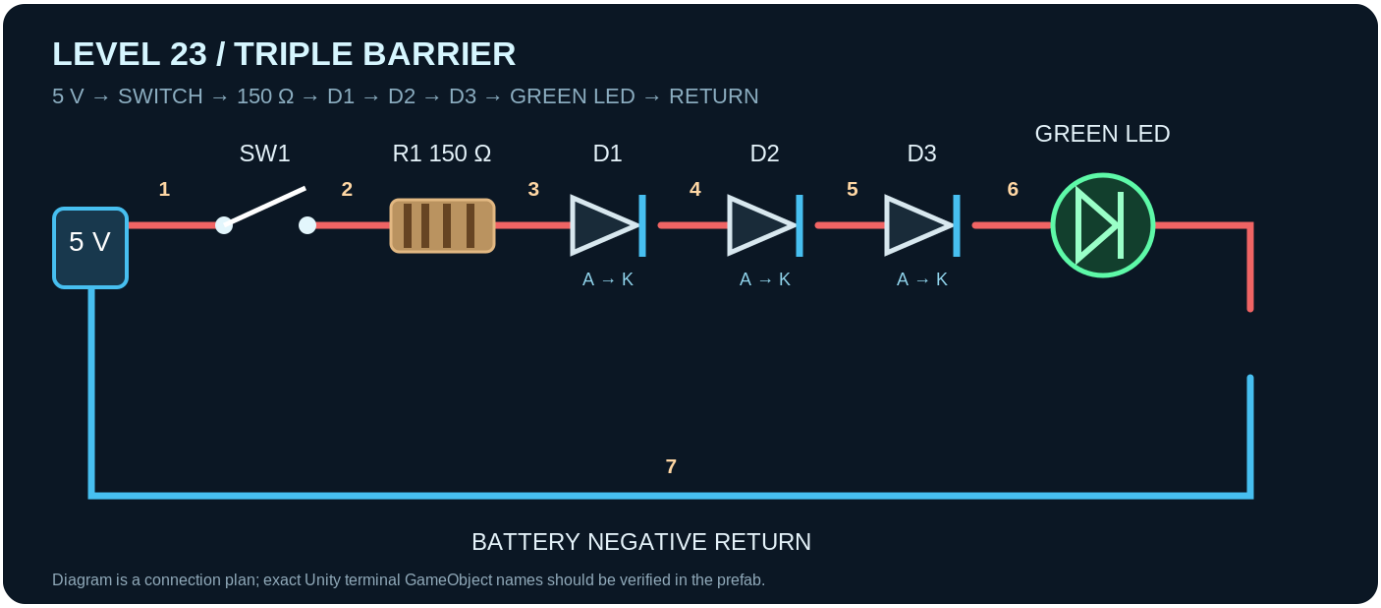
PROJECT SPARK / LEVEL 23

TRIPLE BARRIER - THREE DIODES IN SERIES

Build a 5 V forward-biased series circuit with 3 x 1N4007 diodes, 150 ohm resistor, a power switch, and a green LED. Complete all six objectives. This guide keeps the stable shared solver unchanged.



Build sequence

01 — Backup the working project

Save a copy or commit of the working Level 22 prefab, corrected SparkDiodeSolverDefinition, stable SparkElectricalSolver, and Level 21 transient driver. Do not edit shared solver code for this level.

02 — Duplicate Level 22

In Project view, duplicate Level_22_Layout prefab into Level_23_Layout. Duplicate the Level_22 definition asset into Level_23.asset if not already created. Keep the old assets untouched.

03 — Import Level 23 definition

Select the ORIGINAL Level_23.asset in Unity, then import Level_23_Triple_Diode_Fixed.txt with your editable ScriptableObject importer. Confirm levelId=level_23, displayName=Triple Barrier - Three Diodes in Series, and requiredTargetCount=6.

04 — Link prefab and catalog

On Level_23_Layout > SparkLevelLayout, assign Level_23.asset. On SparkLevelLayoutCatalog, add the new definition/layout pair. Add Level_23.asset to LevelGamePlayManager.levels[] (index 22, for 23 levels). Do not assume catalog registration updates manager array. Use development unlock if needed.

05 — Build components

Set power supply to 5 V. Reuse Power_Switch. Put a 150 Ω resistor in Slot_01, three separate 1N4007 diode items in Slot_02, Slot_03, and Slot_04. The three diode components must be distinct instances. Put the green LED in a fixed scene location (preferred), not a shelf-managed workstation.

06 — Set diode orientation

For every diode, anode must face the previous component, and banded cathode must face the next component/LED. Assign each SparkDiode.AnodeTerminal and CathodeTerminal to the corresponding SparkTerminal objects. The LED anode faces D3 cathode; LED cathode faces battery negative.

WORKSTATIONS, BINDINGS AND WIRING

07 — Configure workstation bindings

On Slot_01/02/03/04 set SparkShelfSlot Kind=Workstation, matching category, occupied-at-start if preinstalled, and appropriate RuntimeComponentId/RuntimeTerminalAId/RuntimeTerminalBId. Set SparkShelfItem category and preferred workstation correctly. Ensure bridge Smart Slots setting agrees with the slot approach; do not mix conflicting runtime IDs.

08 — Configure fixed LED scene bindings

In SparkLevelSceneBindings, bind component ID led to the actual SparkLED component and terminal IDs LED_Anode and LED_Cathode to its real terminals. These are the IDs in the currently exported Level 23 definition. Do not add an empty LED workstation rule that tries to overwrite them.

09 — Connect seven external wires

Use the seven connections in the wiring checklist. The physical wires bridge adjacent components; the component solver definitions handle conduction internally. Verify each connection in SparkCircuitSystem, not only by glow.

10 — Verify solver definitions

Keep the already corrected diode Stamp() that calls AddForwardDrop(anode, cathode, ForwardVoltage, resistance) exactly once; AddForwardDrop internally stamps conductance. Confirm each diode uses the registered SparkDiodeSolverDefinition, and LED uses the registered LED solver definition.

Workstation runtime IDs

Slot	Component ID	Terminal A / B
Slot_01	R_LIMIT_Component	R_LIMIT_Input / R_LIMIT_Output
Slot_02	D_SERIES_1_Component	D_SERIES_1_Anode / D_SERIES_1_Cathode
Slot_03	D_SERIES_2_Component	D_SERIES_2_Anode / D_SERIES_2_Cathode
Slot_04	D_SERIES_3_Component	D_SERIES_3_Anode / D_SERIES_3_Cathode

Fixed LED component ID: led | terminal IDs: LED_Anode and LED_Cathode. Match your Level 23 definition exactly.

Seven external wires

#	From	To
1	Battery_positive	SwitchInput
2	SwitchOutput	Register_Input
3	Register_Output	Diode1_Anode
4	Diode1_Cathode	Diode2_Anode
5	Diode2_Cathode	Diode3_Anode
6	Diode3_Cathode	LED_Anode
7	LED_Cathode	Battery_negative

EVALUATION AND PLAY-MODE CHECKS

11 — Verify bindings and current

In Play Mode press Synchronize + Report Now and confirm all 12 workstation IDs bind (3 resistor + 9 diodes). Confirm fixed LED resolves directly. Close switch and run the solved-source-current audit. Test with all components connected.

12 — Tune and complete

Check 6/6 objectives. Record measured diode drops and resistor current; if necessary adjust only Level_23.asset thresholds, not the shared solver. Confirm Level 22 remains 4/4 and Level 21 charges normally.

Six completion targets

Target ID	Type	Requirement
level23_supply	VoltageRange	4.8–5.2 V
level23_diode1	VoltageRange	0.45–1.0 V
level23_diode2	VoltageRange	0.45–1.0 V
level23_diode3	VoltageRange	0.45–1.0 V
level23_led_on	LEDOn	ON
level23_safe_current	CurrentRange	4.5–6.5 mA

Electrical theory

Approximate series current: (5 V - 3 x 0.7 V - 2.1 V) / 150 ohm = 5.33 mA. Resistor drop approximately 0.8 V. These are estimates; tune thresholds after an actual solver audit.

Pass/fail checklist

[ ] All 12 replaceable-workstation bindings resolve
[ ] Fixed LED resolves via scene bindings
[ ] Seven real SparkCircuitSystem external connections exist
[ ] All three diodes face forward; switch closed
[ ] Electrical solver succeeds; source current is positive
[ ] No short circuit or open return; all six targets pass
[ ] Levels 21 and 22 still operate correctly

Common failure modes

UNBOUND D_SERIES_*: check preinstalled SparkShelfItem occupant, slot/runtime IDs, bridge Smart Slots mode, and actual diode terminal names.
LED UNBOUND: fixed binding IDs `led`, `LED_Anode`, `LED_Cathode` must match Level_23.asset. Do not rely on an empty LED workstation.
Source current 0 A / ClosedReturn NO: check seven real SparkCircuitSystem connections, switch closed, diode/LED polarity, and success of nonlinear solve.
Solver nonconvergent when diode connected: confirm diode model calls `AddForwardDrop(..., onResistance)` once (not conductance) without separate AddConductance.
Level not loading: check `LevelGamePlayManager.levels[]` index 22 and progression access, separately from layout catalog.
Performance: keep the stable dirty-based solver; no global continuous tick; gate debug messages.
Teaching drawing: native `SparkLevelDrawingAsset` creator can be authored as a separate Level 23 asset with overview, wiring, forward drops and target pages; this guide does not claim to generate that Unity asset.